Linear Algebra - HCMUS - April 24th, 2024

Semester 2 : Year 2024 - 2025 - Time Duration: 90 minutes

Question 1. (3 points) Do the following requests

(1) Let A and B be matrices such that $A^T = \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}$ and $B^T = \begin{pmatrix} -1 & 1 \\ 0 & -2 \\ 2 & 1 \end{pmatrix}$

(a) Calculate $(2I_2^{2025} - A)B + (B^T A)^T$

(b) Solve the equation $A(X - B) = B$

(2) Given A, B and $AB - I_n$ are nth level square matrices. Show that if $A, B, AB - I_n$ are invertible, then $A^{-1} + (B^{-1} - A)^{-1}$ is also invertible and find its inverse form.

Question 2. (3 points) For each pairs $(x, y) \in \mathbb{R}^2$, denoting $A_{x,y} = \begin{pmatrix} -8 & -x - 2y - 10 & -4x + y - 7 \\ 2 & 4 & -y + 2 \\ 8 & 2y + 8 & 4x + 8 \end{pmatrix}$

(a) Find the rank of $A_{-2,4}$

(b) Calculate x, y so that $\det A_{x,y}$ reaches minimum value.

(c) Find x such that $A_{x,4}$ is row-wise equivalent to $\begin{pmatrix} 2 & 2 & -7 \\ -4 & -6 & 9 \\ 3 & 4 & -8 \end{pmatrix}$

Question 3. (2 points) In the Oxyz space, for each $\lambda \in \mathbb{R}$, set $(P) : \lambda x - \lambda y + z + 3\lambda = 0$, $(Q_\lambda) : x + 2\lambda y + \lambda z - 5\lambda - 1 = 0$, and $(R_\lambda) : 2\lambda x - y + \lambda z - \lambda + 2 = 0$. Show that for all $\lambda \in \mathbb{R}$, three planes $(P_\lambda), (Q_\lambda), (R_\lambda)$ intersects at only a point and lies on a stable plane as λ changes.

Question 4. (2 points) Let α be a real number different from 0 and

$$A = \begin{pmatrix} 0 & \alpha & \alpha^2 & \alpha^3 & \cdots & \alpha^{2023} & \alpha^{2024} \\ \frac{1}{\alpha} & 0 & \alpha & \alpha^2 & \cdots & \alpha^{2022} & \alpha^{2023} \\ \frac{1}{\alpha^2} & \frac{1}{\alpha} & 0 & \alpha & \cdots & \alpha^{2021} & \alpha^{2022} \\ \frac{1}{\alpha^3} & \frac{1}{\alpha^2} & \frac{1}{\alpha} & 0 & \cdots & \alpha^{2020} & \alpha^{2021} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ \frac{1}{\alpha^{2023}} & \frac{1}{\alpha^{2022}} & \frac{1}{\alpha^{2021}} & \frac{1}{\alpha^{2020}} & \cdots & 0 & \alpha \\ \frac{1}{\alpha^{2024}} & \frac{1}{\alpha^{2023}} & \frac{1}{\alpha^{2022}} & \frac{1}{\alpha^{2021}} & \cdots & \frac{1}{\alpha} & 0 \end{pmatrix}$$

(a) Calculate A^k with k is a positive integer.

(b) Prove that A is invertible and find it's inverse matrix.

END